



Python Variables & Data Types

Data with Dylan | Guided Notes | Python Fundamentals #1

Name: _____

Date: _____

 **HOW TO USE THESE NOTES:** Watch the video and fill in the blanks as Dylan explains each concept. Use the writing boxes to capture your own notes and examples. Complete the practice problem at the end!

Section 1: What Is a Variable?

Topic: Variables — the core building block of Python

What is a variable?

Complete the definition:

A variable is like a _____ that stores _____.

It can hold different kinds of information — fill in the examples Dylan gives:

- Numbers: _____
- Text: _____
- True/False values: _____

Complete the code below:

```
my_num = _____ # stores the number 5
To display the value on screen, use:
_____ (my_num)
```

In your own words — what does the `print()` function do?

Section 2: Naming Your Variables

Topic: Valid variable names & Python naming conventions

Rules for naming variables

Variable names **cannot** contain a _____ — it will throw an error.

For multiple words, the Python convention is to use a _____ between them.

Example: `my_score` ☒ `my score` ☐

There is another naming style (less common in Python):

It is called _____ case. Example: `myScore`

Dylan recommends sticking with: _____

 Circle the **VALID** variable names below:

first name	first_name	1stname	firstName	score1
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Section 3: Integers & Floats

Topic: The first two data types — int and float

Definitions

An **integer (int)** is any _____ or _____ whole number.

A **float** is any number that has a _____.

Key rule to remember: dividing two integers with / always returns a _____.

Operations — fill in the table as Dylan covers each one

CODE	OPERATOR NAME	RESULT / NOTES

In your own words — what does the % (modulo) operator return?

Dylan mentions one common use for modulo — what is it?

Section 4: Strings

Topic: Working with text in Python

What is a string?

A string is any _____ or _____ wrapped in quotes.

You can use "double" or 'single' quotes — just be _____.

What happens if you mix quote types in a string?

String Concatenation

Joining strings together is called _____. You use the + operator.

Fill in the missing part of the code below:

```
first_name = "Dylan"
last_name = "Song"
print(first_name + _____ + last_name)
```

Why do we need the middle part? _____

Section 5: Type Casting

Topic: Converting between data types

What is type casting?

Type casting means converting one data type into _____.

You **must** convert before mixing types in a `print()` statement — otherwise you get a _____ Error.

Fill in the correct type conversion function below:

```
age = 19
print("Dylan is " + _____(age))
```

Does type casting permanently change the variable's type? Circle one and explain why:

YES — it changes permanently

NO — only for that one line

Explain your answer:

Section 6: Booleans

Topic: The final data type — True and False

What is a Boolean?

A boolean can only hold one of two values: _____ or _____.

You store them like any other variable: `my_bool = _____`

When will you most often use booleans?

Dylan teases a future video topic here — what is it?

Section 7: The 4 Main Data Types — Fill In the Table

Topic: Putting it all together

Using what you have learned, complete the full data types reference table:

TYPE	WHAT IT STORES	EXAMPLE	YOUR OWN EXAMPLE
<code>int</code>			
<code>float</code>			
<code>str</code>			
<code>bool</code>			

Section 8: Quick Recap

Test yourself — try to complete these without looking back!

Complete each sentence from memory:

1. A variable is like a _____ that stores _____.
2. The Python naming convention for multi-word variables is: _____
3. The 4 main data types are: _____
4. To join two strings together, you use the _____ operator.
5. Before mixing a number and a string in a `print()` statement, you must: _____

Practice Problem

 **Challenge — Pause the video and try this yourself before Dylan reveals the answer!**

```
name = "Dylan"      height_inches = 67
```

Goal: Print "Dylan is 5 feet 7 inches tall"

To get the feet, I will use the _____ operator. Why? _____

To get the inches, I will use the _____ operator. Why? _____

Write your full print statement below:

```
print( _____ )
```

How did it go? What part tripped you up?

Personal Notes

Use this space to capture any additional observations, questions, or ideas from the video.

Key Terms Reference

Use this table to build your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Variable	
Integer (int)	
Float	
String (str)	
Boolean (bool)	
snake_case	
camelCase	
Concatenation	
Type casting	
str()	
int()	
print()	
Integer division (//)	
Modulo (%)	
TypeError	